

Response surface alignment measure

1 Set-up

AOD at the pair of gridboxes at longitude -134.0625 and latitude -21.875 and longitude -171.5625 and latitude 51.875 (shown in green and blue respectively in Figure 1) are considered, for July.

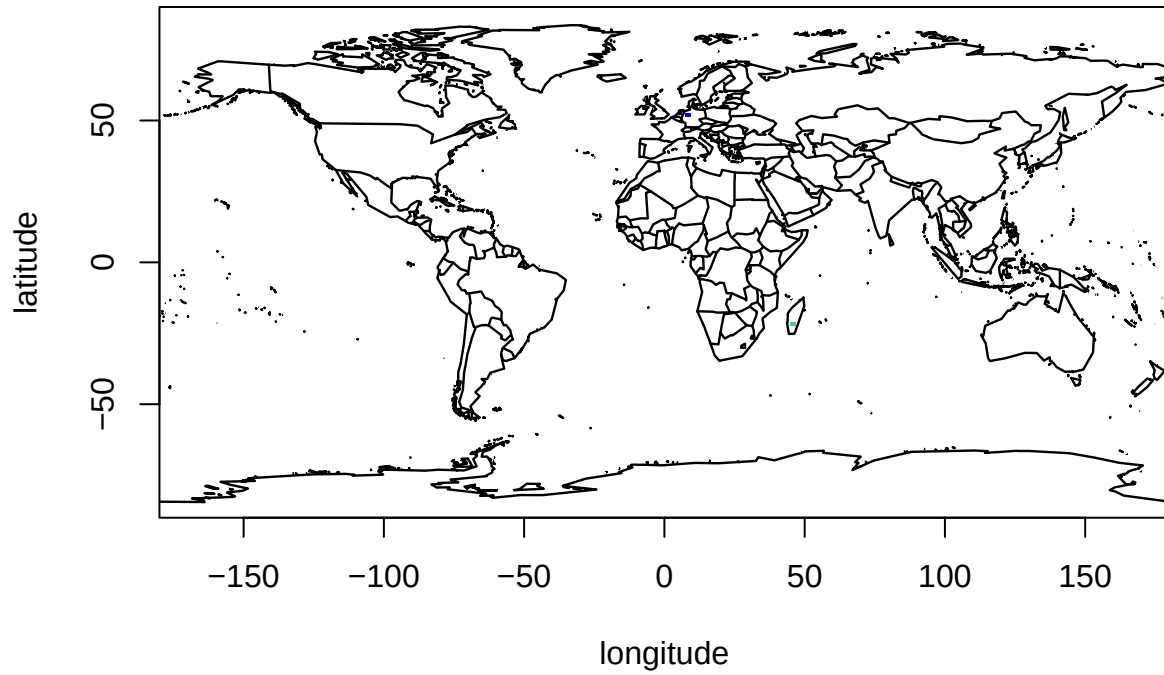


Figure 1: Selected gridboxes highlighted in green and blue.

2 Sensitivity analysis

GAM-based sensitivity analysis is performed for the two outputs. To reach 120% of the variance of the two outputs requires 37 and 37 inputs respectively (with main effects summing to 115% and 115%); see Table 1 and Figure 2..

Table 1: Post-SA main effects for the first (left) and second (right) outputs.

Input	Main effect (%)	Input	Main effect (%)
sea_spray	32	dry_dep_acc	42
dry_dep_acc	24	prim_so4_diam	20
bvoc_soa	22	bvoc_soa	13
kappa_oc	10	kappa_oc	9
prim_so4_diam	3	sea_spray	6
bparam	3	dry_dep_so2	4
bc_ri	2	anth_so2_eur	3
carb_ff_diam	2	bparam	3
dbstdtbs_turb_0	2	dbstdtbs_turb_0	2
oxidants_oh	2	anth_so2_chi	2
volc_so2	2	c_r_correl	2
rain_frac	1	bc_ri	2
autoconv_exp_lwp	1	oxidants_oh	1
anth_so2_nam	1	dry_dep_ait	1
dms	1	ai	1
c_r_correl	1	carb_ff_diam	1
conv_plume_scav	1	prim_moc	0
anth_so2_chi	1	ait_width	0
ait_width	1	autoconv_exp_lwp	0
prim_moc	1	anth_so2_asi	0
sig_w	1	conv_plume_scav	0
dry_dep_so2	1	volc_so2	0
two_d_fsd_factor	1	dms	0
m_ci	0	a_ent_1_rp	0
dry_dep_ait	0	oxidants_o3	0
bl_nuc	0	two_d_fsd_factor	0
anth_so2_eur	0	m_ci	0
carb_bb_diam	0	carb_bb_diam	0
oxidants_o3	0	cloud_ph	0
cloud_ice_thresh	0	anth_so2_nam	0
autoconv_exp_nd	0	cloud_ice_thresh	0
anth_so2_r	0	bl_nuc	0
ai	0	rain_frac	0
a_ent_1_rp	0	autoconv_exp_nd	0
carb_res_diam	0	sig_w	0
cloud_ph	0	anth_so2_r	0
anth_so2_asi	0	carb_res_diam	0

An emulator will be fitted for the union of inputs from Table 1, namely the 37 inputs:

1. sea_spray
2. dry_dep_acc



Figure 2: The variables with main effects summing to at least 95% for (left) output 1 and (right) output 2.

3. bvoc_soa
4. kappa_oc
5. prim_so4_diam
6. bparam
7. bc_ri
8. carb_ff_diam
9. dbsdtbs_turb_0
10. oxidants_oh
11. volc_so2
12. rain_frac
13. autoconv_exp_lwp
14. anth_so2_nam
15. dms
16. c_r_correl
17. conv_plume_scav
18. anth_so2_chi
19. ait_width
20. prim_moc
21. sig_w
22. dry_dep_so2
23. two_d_fsd_factor
24. m_ci
25. dry_dep_ait
26. bl_nuc
27. anth_so2_eur
28. carb_bb_diam
29. oxidants_o3
30. cloud_ice_thresh
31. autoconv_exp_nd
32. anth_so2_r
33. ai
34. a_ent_1_rp
35. carb_res_diam
36. cloud_ph
37. anth_so2_asl

3 Emulator fit and validation

To fit the emulator, we assume a linear mean function of the form

$$m(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p.$$

The Gauss (squared exponential) kernel is used for the covariance function, maximum posterior mode estimation for the parameters estimates (trend, range, SD and the nugget term), and the **RobustGaSP** package to fit the emulators.

For validation of the emulators fitted, as well as calculation of the alignment measure, 100 new LHC points are generated and predictions made at these points. The mean width of the 95% CI's for these predictions are shown in Table 3, along with the mean width of the LOO CV prediction 95% CI's. Note that the range of output 1 is 0.2 and of output 2 is 0.6.

Table 2: Validation checks

Output	Prediction 95% CI mean width	LOO CV 95% CI mean width	% validated in LOO CV
1	0.06	0.06	95
2	0.23	0.23	95

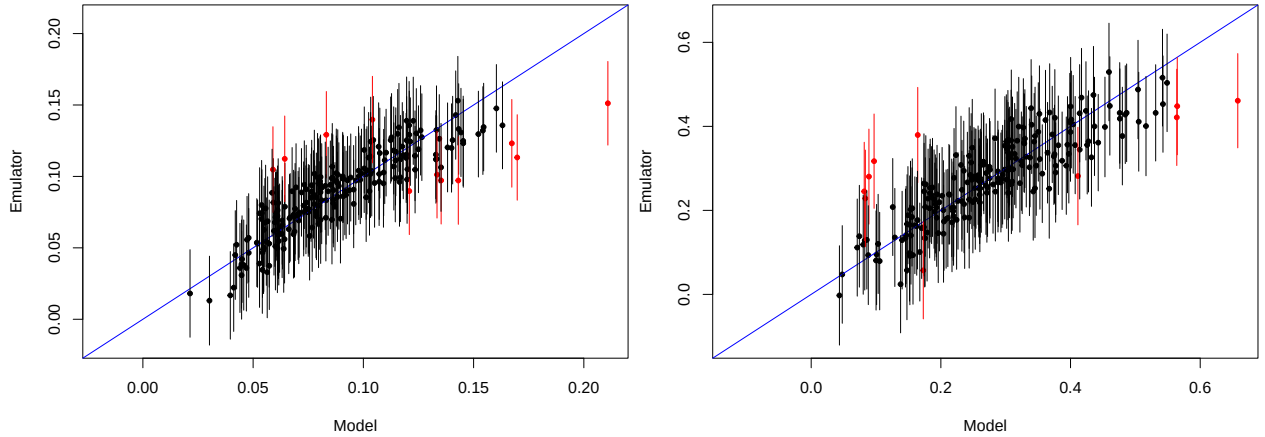


Figure 3: LOO CV validation plot for output 1 (left) and output 2 (right).

4 Check of by-hand calculations

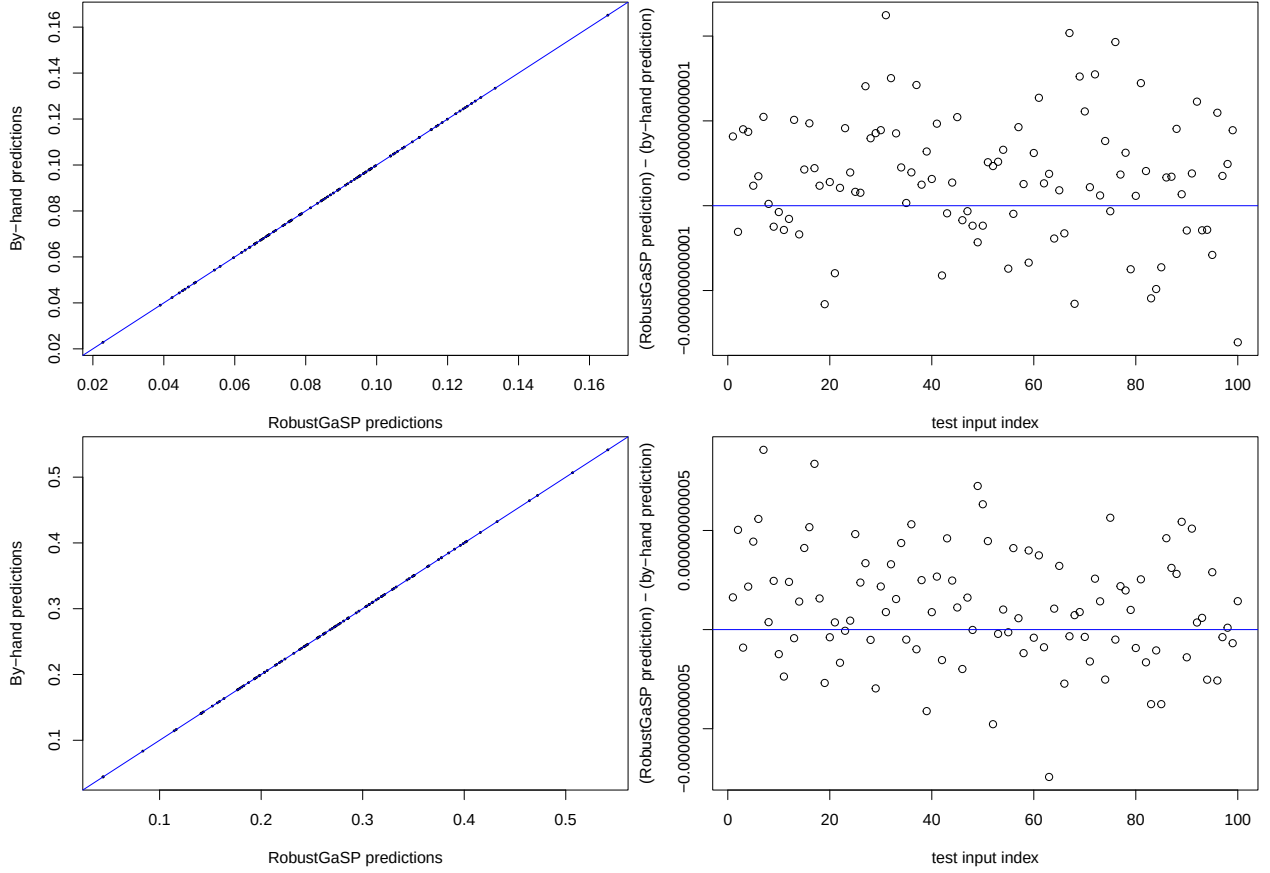


Figure 4: The predictions found by-hand match well with those made by the RobustGaSP package (left), with the latter minus the former found to be essentially zero (right). Top: Output 1. Bottom: Output 2.

Ahead of performing by-hand calculations of the partial derivatives at the testing points, by-hand calculations for the predictions made at these points are plotted against the predictions returned by the **RobustGaSP** package (left hand plots in Figure 3) and the difference between the two plotted against the test input index (right hand plots in Figure 3). Finally, for both we calculate

$$\frac{\sum_{i=1}^N \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{N},$$

where N is the number of test points to get 0 for output 1 and 0 for output 2.

5 Calculation of the alignment measure

Having calculated the emulated partial derivatives for use in the alignment measure, and normalising these, we calculate the AM, getting a value of 0.816.